

Multiscale Contrastive Learning for Domain Generalization: A Reproduction Study with a Corrected Objective, Multi-Seed Evaluation, and Representation-Level Leakage Analysis

Madhusudan Reddy.C
IIIT Sricity

Abstract—Multiscale and Multilayer Contrastive Learning (M2-CL) combines multiscale intermediate feature extraction (M2) with a supervised contrastive objective for domain generalization (DG). We report an independent reproduction on PACS under a compute-constrained setting (four extraction tap locations rather than the original thirteen, ResNet-18 backbone), and use it to ask two questions the original work does not directly address: whether M2-CL training results in greater linearly-decodable domain information relative to an ImageNet-pretrained control, and whether an implementation issue found during this reproduction affects either classification accuracy or that leakage finding. Our initial single-seed reproduction shows M2-CL underperforming an ERM baseline on Sketch, reversing the trend reported in the original paper. We trace part of this to a loss-normalization error (mean- rather than sum-reduction over classes and layers) that scaled down the contrastive term by a factor on the order of the number of classes times the number of layers per batch (approximately $28\times$ when all seven PACS classes are represented in a batch and four layers are averaged; the realized factor varies with batch composition). Because the objective was re-tuned after the fix, the corrected configuration uses a smaller α than the original default; we distinguish the raw loss-scaling error from this subsequent hyperparameter change throughout. After correcting the reduction and, in the same intervention, extending the extraction architecture from four to six taps, a three-seed re-evaluation observes higher accuracy for the corrected model than a comparably re-measured three-seed ERM baseline on three of four domains and by 0.94pp on macro-average; with only three matched seeds per domain, an exact two-sided sign test and a sign-flip permutation test both have a minimum attainable p -value of 0.25, so we report these differences descriptively, and the joint change in loss scaling and tap count prevents attributing them to either intervention individually. We then probe this corrected checkpoint family for domain leakage using the same protocol as the original checkpoints, extending coverage to Sketch for the first time. None of the resulting twelve domain-by-seed comparisons against an ImageNet-only control survive correction for multiple comparisons. We additionally establish an empirical noise floor for the probe itself: repeated measurements of the identical, deterministic ImageNet-only control differ by up to 0.8pp through probe-training randomness alone, which is the

same order as every effect the leakage analysis reports and which strengthens the interpretation that effects of this magnitude are difficult to distinguish from probe-training variability under our protocol.

Index Terms—domain generalization, contrastive learning, PACS, representation probing, reproducibility, ImageNet pre-training

I. INTRODUCTION

Domain generalization (DG) methods aim to train image classifiers that retain accuracy on domains not seen during training. A substantial body of DG work, including contrastive and invariant-representation-learning methods, motivates itself by arguing that a trained model’s internal representations should be pushed toward discarding domain-correlated information while retaining class-relevant information [1]–[3]. This motivation is usually argued informally, and rarely tested by directly measuring how much domain identity remains linearly decodable from a model’s representations at different stages of training.

This paper documents two connected pieces of work built around an independent reproduction of Multiscale and Multilayer Contrastive Learning (M2-CL) [1] on the PACS benchmark [4], which comprises 9,991 images spanning seven object categories across four domains (Photo, Art Painting, Cartoon, and Sketch). First, we reproduce the method under a reduced-capacity, compute-constrained setting, and discover a discrepancy in the original paper’s stated result: M2-CL, which outperforms ERM on every PACS domain in the original paper’s reported results [1], in our single-seed reproduction instead makes performance on Sketch *worse* than a plain ERM baseline. Tracing this discrepancy leads us to a loss-normalization bug in our own implementation, whose correction – combined with a modest architectural change – forms the second part of this work: a corrected, multi-seed re-evaluation. Second, we use both the original (uncorrected) and the corrected checkpoints to ask a question orthogonal to classification accuracy: does M2-CL training result in greater

linearly-decodable domain identity from a model’s frozen representations, relative to an ImageNet-pretrained backbone that receives no PACS data during representation training?

The leakage analysis was initially conducted on the original checkpoints and was subsequently repeated after the implementation correction, to determine whether the resulting null finding persisted under a corrected, multi-seed configuration whose contrastive term has been restored to its intended loss scaling. We report the work in this order – reproduction, then correction, then re-tested leakage analysis – because the sequence itself is scientifically informative: it demonstrates that a null result obtained with an under-scaled training objective can be usefully re-tested once that objective was restored to its intended loss scaling and the corrected configuration had been re-evaluated for classification, rather than assumed to generalize.

Our contributions are:

- An identification and correction of a loss-scaling implementation error that reduced the effective magnitude of the M2-CL contrastive term by approximately $28\times$ when all seven PACS classes are represented and four layers are averaged (the realized factor varying with batch composition), relative to its intended contribution, together with the classification-accuracy consequences of this correction.
- A three-seed comparison between ERM and corrected M2-CL across all four PACS domains, with ERM and M2-CL matched by seed-level source-domain train/validation partitions, providing a stronger basis for comparison than the original single-seed measurements – reported alongside an exact sign-permutation test that makes explicit how little three seeds can support statistically.
- An extension of a source-only, majority-baseline-referenced, paired-bootstrap domain-leakage test to the corrected M2-CL configuration, closing a coverage gap (Sketch) that a loss-scaling artifact had previously left untested.
- An explicit statement of which claims in this paper are, and are not, supported by a clean ablation, given that our architectural and loss corrections were introduced together rather than in isolation.

II. RELATED WORK

Gulrajani and Lopez-Paz [5] showed, under the DomainBed evaluation protocol, that plain ERM matches or exceeds many specialized DG algorithms on PACS and related benchmarks once evaluation protocol is controlled for. Our leakage analysis is complementary to this finding rather than a restatement of it: DomainBed shows that DG methods do not reliably improve downstream accuracy over ERM; the representation-probing analysis in this paper asks whether DG training objectives measurably change how much domain identity is linearly decodable from learned representations in the first place, which is one candidate mechanistic explanation for that empirical finding.

SelfReg [2] and DomCLP [3] both motivate contrastive DG objectives, in part, by arguing that InfoNCE-style losses with cross-domain negative sampling suppress domain-irrelevant common features while amplifying domain-relevant ones, and can therefore introduce or worsen domain leakage if not

carefully constrained. The present work does not directly test either method, but tests the directly related M2-CL objective [1], which uses a within-batch supervised contrastive loss with same-class positive pairs over multiscale extracted representations.

Linear probing [6] is used throughout this work to measure whether domain identity is linearly recoverable from frozen representations, without assuming a particular downstream use of that information.

III. BACKGROUND: M2 AND M2-CL

M2 [1] augments a convolutional backbone (here, ResNet-18 [7]) with *extraction blocks* attached at intermediate depths. Each extraction block applies one or more parallel *concentration pipelines* – a 1×1 convolution, spatial dropout, adaptive max pooling at a fixed output resolution, and a small MLP – to the block’s input feature map, at several pooling resolutions simultaneously. The outputs of all extraction blocks, together with the backbone’s own global-average-pooled final-layer features, are concatenated and passed to a single linear classification head. In the original design, thirteen intermediate convolutional locations are tapped across the backbone [1]. The published description further states that each location feeds two or three parallel concentration pipelines; because it does not specify how many of the thirteen locations belong to each group, the exact total number of pipeline outputs is not recoverable from the published description.

M2-CL adds a supervised contrastive loss computed on the extraction-block outputs. For a batch of N_b samples and layer l , with normalized representation $u_i^{(l)}$ for sample i , the original paper defines

$$p^{(l)}(c) = \frac{\sum_{i=1}^{N_c} \sum_{j=i+1}^{N_c} \mathbf{1}[y_i = y_j = c] e^{u_i^{(l)\top} u_j^{(l)} / \tau}}{\sum_{k=1}^{N_b} \sum_{m=k+1}^{N_b} e^{u_k^{(l)\top} u_m^{(l)} / \tau}} \quad (1)$$

where N_c is the number of batch elements of class c and τ is a temperature hyperparameter. The per-layer loss is $\mathcal{L}^{(l)} = -\sum_{c=1}^C \log p^{(l)}(c)$, and the total training objective is

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \alpha \sum_{l=1}^M \mathcal{L}^{(l)} \quad (2)$$

with $\mathcal{L}_{\text{CE}}$ the standard cross-entropy loss, M the number of layers considered, and α a hyperparameter. The original paper reports default values $\alpha = 0.01$, $\tau = 1.0$, batch size 128, SGD with learning rate 0.001, and 30 training epochs on each PACS fold [1].

IV. INITIAL REPRODUCTION

A. Deviations from the Original Setup

Our initial reproduction used a ResNet-18 backbone with the same batch size (128), optimizer family (SGD), and epoch budget (30) as [1], but differed in three respects, imposed by available compute:

- **Extraction taps.** We attached extraction blocks at the output of each of the four ResNet-18 stages (`layer1`–`layer4`), yielding ten concentration-pipeline outputs plus

TABLE I
INITIAL SINGLE-SEED REPRODUCTION VS. PAPER-REPORTED ACCURACY (%)

Model	Src.	Art	Cart.	Photo	Sketch
ERM	Ours	73.14	70.69	94.37	66.51
	Paper	79.68	77.76	88.91	75.72
M2	Ours	74.27	67.36	95.39	61.06
	Paper	79.86	77.91	95.65	77.90
M2-CL	Ours	74.22	68.30	95.45	61.03
	Paper	81.66	78.42	97.00	77.07

the backbone’s own GAP feature (1792-d total). The original design uses thirteen intermediate tap locations [1]; each location feeds two or three parallel concentration pipelines, so the full design has substantially more pipeline outputs than our ten. The published description does not specify how many of the thirteen locations use two versus three pipelines, so we do not infer an exact full-design pipeline-output count.

- **Learning rate.** We used 5×10^{-4} rather than the paper’s stated 10^{-3} . This deviation was never subsequently tested and is treated as an open limitation (Section XI).
- **Seeds.** The initial reproduction of ERM, M2, and M2-CL used a single seed (seed 0) per condition, rather than the three-seed average reported in [1].

B. Initial Results

Table I reports the initial single-seed reproduction alongside the values reported in [1].

Two observations from Table I motivate the remainder of this paper. First, the Photo domain shows the smallest reproduction discrepancy (within 0.3–5.5 percentage points, pp), while Art, Cartoon, and especially Sketch trail the paper by considerably larger margins, consistent with reduced extraction capacity and single-seed variance compounding on the domains most different from ImageNet’s photographic pretraining distribution. Second, and more specifically: in [1], M2-CL exceeds ERM on every domain including Sketch (77.07% vs. 75.72%). In our reproduction, M2-CL instead falls *below* ERM on Sketch by 5.48pp (61.03% vs. 66.51%) – a reversal of the reported trend, and, at the time this reproduction was performed, unexplained.

V. DOMAIN-LEAKAGE INVESTIGATION

Independently of the accuracy discrepancy above, we asked a narrower question: at which stage of a training pipeline is domain identity actually rendered linearly decodable from a model’s representations? A substantial body of DG literature implicitly assumes that if a trained model’s representations still permit domain identity to be linearly recovered, this is attributable to how the model was trained [2], [3]. We test this assumption with a five-condition sequential ablation: randomly initialized ResNet-18 (5 seeds), ImageNet-pretrained ResNet-18 (trained without PACS exposure), ERM, M2, and M2-CL, using the checkpoints from Section IV.

A. Protocol

For each held-out target domain, a linear domain probe is trained on a stratified 70/30 split of the three *source*

TABLE II
ARCHITECTURAL BIAS AND PRETRAINING AMPLIFICATION BY DOMAIN.

Domain	Random-init (mean±std)	Majority	Arch. bias	ImageNet	Pretrain. amp.
Photo	66.2±1.1%	47.2%	+19.0pp	88.3%	+22.1pp
Art Painting	72.0±2.2%	49.5%	+22.5pp	91.2%	+19.2pp
Cartoon	71.1±1.6%	51.4%	+19.7pp	88.7%	+17.6pp
Sketch	44.7±3.8%	38.7%	+6.0pp	66.5%	+21.8pp

domains only; the held-out domain is never used for domain probing, to avoid leaking the DG evaluation signal into the leakage measurement. Because PACS domains are of unequal size, the baseline reported for each split is that split’s own majority-class rate over its specific three-domain source pool (ranging from 38.7% to 51.4%), not a naive uniform baseline. For the paired significance test, a single comparable 512-d backbone-level feature is used across all conditions: the GAP output of the final convolutional block (`backbone_final` for ImageNet-only/ERM, `gap_backbone` for M2/M2-CL, which are architecturally equivalent).

The domain probe itself is a single linear layer (no hidden units) mapping the 512-d feature to the number of source domains, trained with Adam (learning rate 1×10^{-3}) for 30 epochs with batch size 64 and a cross-entropy loss, on raw, unnormalized features. Probe initialization and training use a fixed probe seed shared across every condition within a fold, so an accuracy difference between conditions reflects the backbone features rather than probe-training variance. The 70/30 source-domain split is generated once per fold with a fixed split seed and reused across every condition, so all paired comparisons within a fold share an identical test set. The paired bootstrap test resamples the shared test-set indices with replacement (2000 resamples per test); on each resample it recomputes both conditions’ accuracy on the identical resampled indices and records the difference. The reported p -value is two-sided, $p = 2 \min(\Pr[\Delta^* \leq 0], \Pr[\Delta^* \geq 0])$ over the bootstrap distribution of Δ^* , and the reported 95% interval is the 2.5th/97.5th percentile of that same distribution.

To quantify variability attributable solely to probe training, we repeatedly trained the same linear probe on the identical frozen ImageNet-only features and identical source-domain split across five probe-initialization seeds while holding all other probe settings fixed. Across these repeated measurements, the maximum observed pairwise accuracy difference was 0.8 percentage points. We use this value only as an empirical reference for the scale of probe-training variability under our protocol, rather than as a formal statistical threshold.

B. Architectural Bias and Pretraining Amplification

Table II reports random-initialization domain-probe accuracy (mean ± standard deviation across five weight-initialization seeds) against the majority-class baseline, alongside ImageNet-pretrained accuracy.

The standard deviations across the five initialization seeds range from 1.1 to 3.8pp, indicating that the architectural-bias estimates vary moderately across initialization seeds. Sketch shows a markedly smaller architectural bias than the three photographic domains, consistent with randomly

TABLE III
 PAIRED BOOTSTRAP TEST: ORIGINAL (SINGLE-SEED, FOUR-TAP) DG
 CONDITION VS. IMAGENET-ONLY CONTROL.

Domain	Cond.	Δ (pp)	Raw p	BH-significant
Photo	ERM	+1.40	0.021	No
Photo	M2	+0.64	0.322	No
Photo	M2-CL	+0.80	0.206	No
Art Painting	ERM	+0.38	0.528	No
Art Painting	M2	-0.42	0.526	No
Art Painting	M2-CL	-0.42	0.519	No
Cartoon	ERM	-0.09	0.924	No
Cartoon	M2	-0.17	0.770	No
Cartoon	M2-CL	-0.48	0.385	No

initialized convolutional filters responding less distinctively to line-drawing statistics than to photographic texture; its pretraining-amplification term, however, is comparable to or larger than the other three domains, which complicates a simple underrepresentation-suppresses-leakage account of the Sketch domain.

C. Significance Testing of DG Training Conditions

Table III reports paired bootstrap test results (2000 resamples, Benjamini-Hochberg correction across the nine comparisons) comparing each DG-trained condition against the ImageNet-only control, for the three domains (Photo, Art Painting, Cartoon) with checkpoints available for all three conditions in this initial reproduction. Sketch is excluded from this table because, at this stage of the investigation, no M2-CL checkpoint had been trained with Sketch held out.

None of the nine comparisons remain significant after correction. The single nominally significant raw p -value (ERM, Photo, $p = 0.021$) does not survive Bonferroni correction ($\alpha/9 = 0.0056$) and is consistent with a false positive expected by chance: across these nine nominal comparisons, approximately 0.45 false positives are expected under a simple independent-tests null model at $\alpha = 0.05$. At this stage of the investigation, we noted two limitations explicitly: (i) the three domains available for this test are also the three domains with the smallest reproduction gap relative to [1] (Table I), so the finding cannot yet speak to Sketch, and (ii) the M2-CL checkpoint used here had not yet been checked for the loss-scaling issue described next, so a null leakage result obtained with a possibly under-scaled contrastive objective could, in principle, simply reflect that the objective was not doing much of anything.

VI. CORRECTING THE M2-CL OBJECTIVE

A. The Loss-Scaling Error

Revisiting the training implementation against Eq. (1)–(2), our reproduction’s contrastive-loss implementation reduced $-\sum_c \log p^{(l)}(c)$ to a *mean* over classes present in the batch, and reduced the layer sum $\sum_{l=1}^M$ to a *mean* over layers, rather than the sum specified in Eq. (2). With approximately seven classes and four layers, the class-sum-to-mean change alone reduces the term by a factor of roughly $C \approx 7$, and the layer-sum-to-mean change reduces it by a further factor of roughly $M = 4$; compounded, this reduces the raw magnitude

of the contrastive term by a factor on the order of their product (approximately $7 \times 4 = 28\times$) before the stated α hyperparameter was even applied. Because [1]’s own sensitivity analysis shows α has a substantial effect on downstream accuracy in both directions (moving from near-zero degrades gracefully toward the M2-only baseline, while $\alpha = 1$ can degrade accuracy by up to 20pp), an implemented contrastive term with magnitude approximately $28\times$ smaller than the corresponding sum-reduced objective when all seven classes and four layers contribute provides a plausible explanation for why M2 and M2-CL produced near-identical accuracy in our initial reproduction (61.06% vs. 61.03% on Sketch, Table I): the contrastive term was contributing a much smaller loss signal.

We corrected the implementation to sum, rather than average, over both classes and layers, matching Eq. (2) exactly, and re-swept α against a wider grid than the original single default value.

B. Search Protocol

To select a corrected configuration without using held-out target test labels for model selection, all search decisions in this stage were made using *source-domain validation accuracy only*; the target (held-out) domain’s test accuracy was computed and logged for later analysis but never used to choose among configurations. Screening runs used 15 training epochs (rather than the full 30) and a single seed, with Sketch held out as a fixed proxy target for the search. Sketch was used only as a pre-specified proxy target for development, chosen before the search began because it was the domain exhibiting the original reproduction discrepancy (Section IV); no Sketch test-domain accuracy was used in any model-selection decision, only source-domain validation accuracy.

The search covered: (i) an α grid of $\{0, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$; (ii) an architectural comparison between the original four-tap extraction configuration and a six-tap configuration that additionally taps the output of the first residual sub-block within `layer1` and `layer2`; (iii) an augmentation comparison between the baseline pipeline (random-resized crop, horizontal flip, colour jitter, and a small random-grayscale probability), a stronger-crop variant, and a shape-oriented variant that converts training images to full grayscale; and (iv) a temperature τ grid of $\{0.5, 1.0, 1.5\}$.

We report the search qualitatively rather than as a table of exact intermediate validation figures, for two reasons. First, differences across the α and τ grids were, in every case we observed, on the order of a few tenths of a percentage point on source-domain validation accuracy – differences we regard as within the noise floor of a single-seed, 15-epoch screening run, and therefore not meaningful rankings in themselves. Second, one intervention produced a clear, non-noise signal: the shape-oriented (full-grayscale) augmentation reduced source-domain validation accuracy by approximately 25 percentage points relative to the baseline augmentation, consistent with this intervention removing colour information the three source domains (Art, Cartoon, Photo, all of which retain substantial colour content) genuinely require, and it was excluded on this basis. The six-tap architecture produced a validation accuracy

TABLE IV
SUMMARY OF PRINCIPAL TRAINING CONFIGURATIONS.

Component	ERM	Historical M2-CL	Corrected M2-CL
Backbone	ResNet-18	ResNet-18	ResNet-18
Extraction taps	–	4	6
α	–	0.01	0.003
τ	–	1.0	1.0
Batch size	128	128	128
Epochs	30	30	30
Learning rate	5×10^{-4}	5×10^{-4}	5×10^{-4}
Augmentation	baseline	baseline	baseline
Seeds (final accuracy)	3	1 historical	3

at or slightly above the four-tap configuration, though again by a margin comparable to single-seed noise at the screening stage; we retained it as the final architecture on the basis that it is directionally closer to the extraction capacity of the original design [1] without approaching its full computational cost. The τ grid produced validation accuracies that did not distinguish among 0.5, 1.0, and 1.5 to within the precision we measured; we selected $\tau = 1.0$, matching the original paper’s default, on the basis that a tie should be broken toward the published default rather than an arbitrarily first-encountered value.

The resulting locked configuration is: $\alpha = 0.003$, $\tau = 1.0$, six extraction taps, baseline augmentation, batch size 128, 30 training epochs.

We emphasize explicitly that this locked configuration changes two things relative to the original single-seed M2-CL checkpoint of Section IV simultaneously: the loss scaling, and the number of extraction taps. No experiment in this paper trains a six-tap model without the contrastive loss (i.e., a tap-matched M2 control). Any classification-accuracy change reported below for this configuration is therefore attributable to the corrected configuration as a whole, and should not be interpreted as an isolated causal effect of either the loss fix or the additional taps individually.

VII. EXPERIMENTAL CONFIGURATION SUMMARY

Table IV summarizes the principal training configurations used in the study. The historical M2-CL condition refers to the original four-tap implementation, whereas the corrected M2-CL condition uses the locked six-tap configuration described in Section VI.

VIII. CORRECTED, MULTI-SEED RE-EVALUATION

A. Final M2-CL Results

The locked configuration was evaluated with the standard leave-one-domain-out protocol across all four PACS domains, with three random seeds per domain (30 epochs each), for a total of twelve training runs. Table V reports the resulting target-domain test accuracy – the accuracy on the entirely held-out domain, evaluated exactly once per run, and distinct from the source-domain validation accuracy used for model selection in Section VI.

Relative to the single-seed original M2-CL checkpoint (Table I), the corrected configuration increases the Sketch accuracy from the historical single-seed value of 61.03% to a three-seed mean of 63.97% (+2.94pp) and the four-domain average by

TABLE V
FINAL CORRECTED M2-CL (SIX-TAP) TARGET-DOMAIN TEST ACCURACY, 3 SEEDS.

Domain	Seed 0	Seed 1	Seed 2	Mean $\pm$ SD
Art Painting	74.51	76.03	75.29	75.28 $\pm$ 0.76
Cartoon	68.43	73.04	68.05	69.84 $\pm$ 2.78
Photo	95.57	95.69	95.75	95.67 $\pm$ 0.09
Sketch	62.89	65.41	63.60	63.97 $\pm$ 1.30

TABLE VI
MATCHED THREE-SEED ERM BASELINE, TARGET-DOMAIN TEST ACCURACY.

Domain	Seed 0	Seed 1	Seed 2	Mean $\pm$ SD
Art Painting	72.85	74.80	72.22	73.29 $\pm$ 1.35
Cartoon	70.78	67.53	66.08	68.13 $\pm$ 2.41
Photo	94.67	95.63	94.91	95.07 $\pm$ 0.50
Sketch	66.20	61.49	65.77	64.49 $\pm$ 2.60

1.44pp (74.75% $\rightarrow$ 76.19%). The Cartoon domain shows the largest seed-to-seed standard deviation of the four (2.78pp), which we note explicitly rather than report only the mean, since a single favourable seed (73.04%, compared to 68.43% and 68.05% for the other two) would otherwise give an overly optimistic impression of this domain’s stability.

B. A Matched Three-Seed ERM Baseline

The comparison above uses a single-seed original M2-CL checkpoint as one reference point. Because the ERM baseline in Table I was likewise measured with a single seed, we additionally trained ERM with the same three-seed, four-domain protocol, to obtain a baseline comparable in seed count to the corrected M2-CL evaluation. Table VI reports the result.

Notably, the three-seed ERM mean on Sketch (64.49%) is lower than the single-seed value originally reported in Table I (66.51%), by 2.02pp. This indicates that the original single-seed ERM measurement was itself a comparatively favourable draw for that domain, and is a useful illustration, independent of any DG-method comparison, of why single-seed measurements on PACS should be interpreted cautiously, particularly on the Sketch domain, which shows the largest seed-to-seed variability of the four in both Tables V and VI.

C. Final Matched Comparison

Table VII places all three reference points side by side: the historical single-seed ERM and M2-CL measurements from Section IV, and the matched three-seed measurements from this section. Figure 1 visualizes the same values.

The seed-level differences in Table VIII show agreement in sign across all three seeds for Art Painting and Photo, but mixed signs for Cartoon and Sketch. With only three matched seeds per domain, an exact two-sided sign-permutation test has $2^3 = 8$ possible sign assignments and a minimum attainable p -value of $2/8 = 0.25$, even when all three differences share the same sign; we therefore report these differences descriptively rather than as a formal superiority claim.

For all four-domain averages reported in this section, we use the macro-average across target domains,

$$\text{AvgAcc} = \frac{1}{4} \sum_{d \in \{\text{Art}, \text{Cartoon}, \text{Photo}, \text{Sketch}\}} \text{Acc}_d. \quad (3)$$

TABLE VII

FINAL MATCHED COMPARISON. ERM AND CORRECTED M2-CL ARE BOTH THREE-SEED MEANS; THE HISTORICAL M2-CL COLUMN IS THE SINGLE-SEED CHECKPOINT FROM TABLE I, RETAINED FOR CONTINUITY WITH SECTION V AND LABELLED ACCORDINGLY. THE MACRO-AVERAGE DELTA (+0.94pp) IS COMPUTED FROM THE UNROUNDED PER-SEED VALUES, NOT BY DIFFERENCING THE ALREADY-ROUNDED ROW MEANS SHOWN HERE.

Domain	ERM (3-seed)	M2-CL (hist., 1-seed)	M2-CL (3-seed)	M2-CL— ERM	Corr. M2-CL vs. paper
Art Painting	73.29	74.22	75.28	+1.99	−6.38
Cartoon	68.13	68.30	69.84	+1.71	−8.58
Photo	95.07	95.45	95.67	+0.60	−1.33
Sketch	64.49	61.03	63.97	−0.52	−13.10
Average	75.24	74.75	76.19	+0.94	−7.35

TABLE VIII

MATCHED SEED-LEVEL CLASSIFICATION DIFFERENCES, CORRECTED M2-CL MINUS ERM (PERCENTAGE POINTS), COMPUTED FROM FULL-PRECISION PER-SEED ACCURACIES AND ROUNDED ONLY AT THE FINAL STEP. SEED INDICES REFER TO IDENTICAL SOURCE-DOMAIN TRAIN/VALIDATION PARTITIONS; MODEL INITIALIZATION AND OPTIMIZATION TRAJECTORIES ARE NOT STRICTLY PAIRED. THE EXACT TWO-SIDED SIGN-PERMUTATION p -VALUE USES THE THREE SIGNED DIFFERENCES ONLY.

Domain	Seed 0	Seed 1	Seed 2	Mean Δ	Exact p
Art Painting	+1.66	+1.22	+3.08	+1.99	0.25
Cartoon	−2.35	+5.50	+1.96	+1.71	1.00
Photo	+0.90	+0.06	+0.84	+0.60	0.25
Sketch	−3.31	+3.92	−2.16	−0.52	1.00

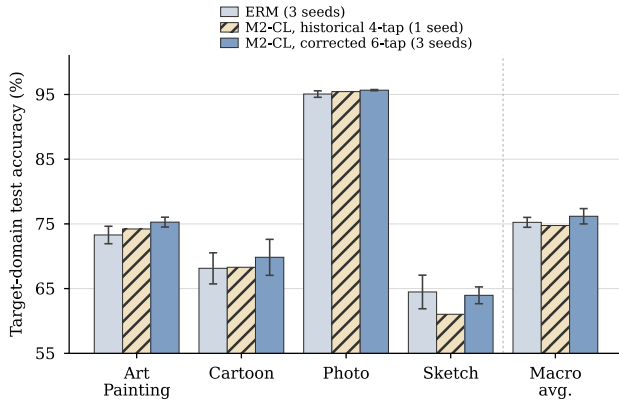


Fig. 1. Target-domain test accuracy by domain, visualizing Table VII. Error bars show ± 1 SD across three seeds where available (ERM and corrected M2-CL); the historical M2-CL checkpoint is a single measurement and carries no error bar.

Thus, each PACS domain contributes equally regardless of its number of test images.

Under the three-seed comparison, the corrected M2-CL configuration exceeds the ERM baseline on three of four domains and on the four-domain average (+0.94pp), with seed-level signs agreeing across all three seeds for Art Painting and Photo (Table VIII). We do not treat this as a formal superiority claim: with only three matched seeds, the exact sign-permutation test cannot reach $p < 0.05$ regardless of outcome, so these differences are reported descriptively. The Sketch difference (−0.52pp, mixed-sign across seeds) is treated as inconclusive rather than as evidence of a disadvantage. What we can say is that the clear reversal observed in the initial

single-seed reproduction (Table I) does not persist once both conditions are measured to the same three-seed standard. This comparison reflects the corrected configuration as a whole and does not isolate the loss correction from the additional extraction taps (Section VI). The reproduction remains 1.3–13.1pp below the values reported in [1] on every domain, most substantially on Sketch, consistent with reduced extraction capacity, the untested learning-rate deviation (Section XI), and reduced training-image counts relative to the full protocol used in the original paper.

IX. ROBUSTNESS OF THE DOMAIN-LEAKAGE RESULT

A. Motivation

The null result in Table III was obtained using the original, single-seed M2-CL checkpoint, which Section VI shows carried a contrastive loss term contributing negligible effective magnitude relative to its intended value. A reviewer could reasonably ask whether this null result reflects a genuine property of the M2-CL objective, or simply reflects that the objective, as implemented, was not doing much of anything. Table V shows that the corrected, six-tap configuration produces a different classification outcome from the historical checkpoint, including an increase from the historical single-seed Sketch accuracy of 61.03% to the corrected three-seed mean of 63.97% (+2.94pp). This provides a materially different configuration – a corrected configuration that produces a different observed classification outcome – with which to re-test the leakage question, and, as a direct consequence of Table V now covering all four domains, allows the Sketch domain to be probed for the first time in this analysis.

B. Method

We follow the same protocol as Section V: the same 512-d GAP-of-final-layer feature (architecturally identical regardless of extraction tap count, since the additional taps in the six-tap configuration are parallel branches that do not feed back into the backbone’s main forward path), the same source-only 70/30 stratified domain-probe split, and the same paired-bootstrap methodology (2000 resamples) against the ImageNet-only control used in Table III. Because we now have three independently trained checkpoints per domain rather than one, we treat these three seeds as replicate measurements of a single condition rather than as three independent statistical hypotheses: we report per-seed paired-bootstrap p -values individually, and separately report the mean and standard deviation of accuracy and of the accuracy delta across seeds. We apply Benjamini-Hochberg correction across all twelve resulting domain-by-seed comparisons, while noting that this is a conservative choice relative to correcting only across the four domain-level hypotheses.

C. Results

Table IX reports the result for all four PACS domains.

None of the twelve domain-by-seed comparisons remain significant after Benjamini-Hochberg correction. The single smallest raw p -value (Art Painting, seed 0, $p = 0.026$) does

TABLE IX
 PAIRED BOOTSTRAP TEST: CORRECTED SIX-TAP M2-CL (3 SEEDS) VS.
 IMAGENET-ONLY CONTROL, ALL FOUR PACS DOMAINS.

Domain	Seed	Δ (pp)	Raw p	BH-significant
Art Painting	ImageNet-only: 91.19%; M2-CL: 90.57 $\pm$ 0.65%			
	0	-1.30	0.026	No
	1	-0.55	0.377	No
	2	+0.00	0.988	No
Cartoon	ImageNet-only: 88.41%; M2-CL: 88.05 $\pm$ 0.45%			
	0	-0.04	0.984	No
	1	-0.87	0.138	No
	2	-0.17	0.796	No
Photo	ImageNet-only: 88.27%; M2-CL: 89.13 $\pm$ 0.23%			
	0	+0.96	0.125	No
	1	+0.60	0.333	No
	2	+1.04	0.077	No
Sketch	ImageNet-only: 67.29%; M2-CL: 67.82 $\pm$ 0.83%			
	0	+1.37	0.135	No
	1	-0.27	0.807	No
	2	+0.49	0.631	No

not survive correction; across the twelve nominal comparisons, approximately 0.6 false positives are expected under a simple independent-tests null model at $\alpha = 0.05$. We note, however, that the three seeds within a given domain share the same fixed source-domain test split and the same ImageNet-only control measurement, and are therefore not fully independent draws in the statistical sense; the Benjamini-Hochberg procedure’s formal false-discovery-rate guarantee assumes independence or positive dependence among the tests it corrects, so this correction is best read here as a standard, conservative convention for reporting multiplicity rather than as an exact bound. Under either reading, we do not treat the single nominally low p -value as evidence of a domain-specific effect. Sign consistency across seeds is worth reporting descriptively even where individual p -values are not significant: the Art Painting and Cartoon deltas are negative (or at worst 0.00) in all three seeds, Photo is positive in all three seeds with the tightest seed-to-seed spread of any domain (± 0.23 pp), and Sketch shows a sign change across seeds consistent with a clean null result. We report the Photo trend descriptively as a candidate direction for future, higher-powered investigation, and explicitly do not claim it as an established effect given that none of its three individual comparisons reach significance.

D. Interpretation

This result extends, rather than reverses, the finding of Table III: a materially stronger and corrected, multi-seed M2-CL configuration still does not produce domain-probe accuracy distinguishable from an ImageNet-only control on any of the four PACS domains, including Sketch, which had no M2-CL checkpoint available at the time of the original analysis. Combined with Section VIII, this configuration is also, at the time of writing, the only DG-trained condition in this analysis measured across multiple independently trained seeds rather than a single checkpoint – directly addressing what we had previously flagged as the weakest point of the leakage claim, for the M2-CL condition specifically. ERM and M2 remain single-checkpoint measurements in the leakage analysis and are not re-evaluated here.

X. DISCUSSION

Taken together, Sections VIII and IX support a specific and, we think, non-obvious conclusion: a higher observed classification accuracy in this reproduction (the Sketch recovery in Table VII) occurred without an accompanying change in linearly-decodable domain identity (Table IX). If the corrected configuration that changes observed classification accuracy did not also change how much domain information is linearly recoverable from the backbone-level representation, this is consistent with the improvement arising from better class-relevant capacity – plausibly from the additional extraction taps, though this cannot be isolated from the loss correction with the experiments performed here – rather than from a change in the amount of domain information present in the representation the classifier consumes. This is a more specific and more defensible claim than either “M2-CL improves domain generalization” or “M2-CL removes domain information”; both of those broader claims would require evidence this study does not provide.

We also note, as a secondary and more general observation about reproduction methodology, that the initial single-seed ERM measurement in this study (Table I) was itself a favourable-side draw on the Sketch domain relative to its own three-seed mean (Table VI). Part of the accuracy reversal that motivated the loss-scaling investigation in Section VI was therefore attributable not only to the M2-CL implementation, but to single-seed variance in the ERM baseline it was being compared against. We flag this as a general caution for single-seed PACS comparisons, consistent with the seed-to-seed standard deviations of 1.3–2.8pp we observe across both trained conditions in this study.

XI. LIMITATIONS

- **No tap-matched, loss-off ablation.** The corrected configuration in Section VI changes both the contrastive loss scaling and the number of extraction taps relative to the original checkpoint. No experiment in this paper trains a six-tap model without the contrastive loss, so the classification-accuracy change in Table VII cannot be attributed to either factor in isolation. This ablation (six-tap M2, no contrastive term, matched seeds) is feasible future work.
- **Reduced extraction capacity.** Both the original (four-tap) and corrected (six-tap) reproductions use fewer intermediate extraction tap locations than the thirteen used in [1]; the full design further expands those locations into multiple concentration pipelines each. This is a plausible contributor to the remaining 1.3–13.1pp gap to the original paper’s reported accuracy (Table VII) and was not systematically varied beyond the four-vs-six-tap comparison in Section VI.
- **Untested learning-rate deviation.** All experiments in this paper use $\text{lr} = 5 \times 10^{-4}$, rather than the 10^{-3} reported in [1]. This deviation predates and is orthogonal to the loss-scaling correction, and was never itself swept or corrected.
- **Low power of the three-seed classification comparison.** The three-seed ERM–M2-CL comparison (Table VIII) is descriptive rather than formally powered: with three matched seeds, an exact two-sided sign-permutation test has a minimum attainable p -value of 0.25, so the reported differences

should be read as estimates of direction and magnitude rather than significance evidence.

- **Single-checkpoint ERM and M2 leakage measurements.** While Section IX extends the M2-CL leakage measurement to three seeds and four domains, the ERM and M2 conditions in this analysis (Table III) remain single-checkpoint measurements. The paired bootstrap test controls for test-set sampling noise but cannot rule out variation across independently retrained instances of ERM or M2.
- **Sequential ablation, not causal decomposition.** As in Section V, each stage of the five-condition ablation is conditional on the stage before it; the architectural-bias and pretraining-amplification figures in Table II should not be read as independent, additive causal contributions.
- **Linear decodability, not causal use.** Domain-probe accuracy throughout this paper measures whether domain identity is linearly recoverable from a frozen representation, not whether a downstream classifier actually uses domain-correlated features when making its predictions. Relatedly, a null result for a linear probe does not imply that nonlinear domain information is absent from the representation; a probe with greater capacity could in principle recover structure a linear probe cannot.
- **Non-independence among corrected tests.** As noted in Section IX, the three per-domain seeds in Table IX share a fixed test split and control measurement and are not fully independent; the Benjamini-Hochberg correction applied there should be read as a conservative reporting convention rather than a formally exact false-discovery-rate bound.
- **Screening-stage search was not exhaustively logged.** The hyperparameter and architecture search in Section VI used single-seed, 15-epoch screening runs on a single proxy target domain (Sketch); we report its qualitative conclusions rather than a full table of intermediate validation figures, and this search was not independently repeated on other target domains before the configuration was locked.
- **A supplementary layer-wise/class-probe breakdown was prepared but is not reported.** An earlier, separately-implemented probing script was used to compute a more granular layer-by-layer and class-probe breakdown for the original single-run checkpoints of Section IV. That script uses a different train/test-split and probe-training configuration than the one described in Section V, and its domain-probe deltas for the same checkpoints do not agree in sign with Table III in several cells. Because the two scripts are not verified to implement an identical protocol, we do not consider the supplementary breakdown reliable enough to report alongside Table III, and omit it rather than present two disagreeing numbers for the same comparison. Reconciling the two scripts, or regenerating the granular breakdown with the Section V protocol, is left to future work.

XII. CONCLUSION

We reproduced M2-CL [1] under a compute-constrained setting, identified a loss-scaling implementation error that under-weighted its contrastive objective by a factor of approximately 28 $\times$, and corrected it alongside a modest extraction-architecture change. Under a matched three-seed comparison,

the corrected configuration shows higher observed accuracy than a correspondingly re-measured ERM baseline on three of four PACS domains and on average, and no longer shows the Sketch-domain reversal present in our initial single-seed reproduction. Using this corrected, multi-seed checkpoint family, we extended a source-only, paired-bootstrap domain-leakage test – previously limited to three domains and a single, later-found-to-be-under-scaled checkpoint – to all four PACS domains including Sketch. None of the resulting twelve domain-by-seed comparisons against an ImageNet-pretrained control survive correction for multiple comparisons, extending the original null finding to a materially stronger intervention rather than resolving it into a positive effect. We report this alongside the observed classification-accuracy change as a result that was not accompanied by a detectable change in linearly-decodable domain identity.

ACKNOWLEDGMENT

The author thanks Dr. Rajendra Prasath for supervision of the honours research this work is part of.

REFERENCES

- [1] A. Ballas and C. Diou, “Multiscale and multilayer contrastive learning for domain generalization,” *IEEE Trans. Artificial Intelligence*, vol. 5, no. 12, pp. 6253–6266, Dec. 2024, doi: 10.1109/TAI.2024.3377173.
- [2] D. Kim, Y. Yoo, S. Park, J. Kim, and J. Lee, “SelfReg: Self-supervised contrastive regularization for domain generalization,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 9619–9628.
- [3] J.-S. Lee, N.-r. Kim, and J.-H. Lee, “DomCLP: Domain-wise contrastive learning with prototype mixup for unsupervised domain generalization,” in *Proc. AAAI Conf. Artificial Intelligence*, 2025, pp. 18119–18127.
- [4] D. Li, Y. Yang, Y.-Z. Song, and T. M. Hospedales, “Deeper, broader and artier domain generalization,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 5543–5551.
- [5] I. Gulrajani and D. Lopez-Paz, “In search of lost domain generalization,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.
- [6] G. Alain and Y. Bengio, “Understanding intermediate layers using linear classifier probes,” *arXiv preprint arXiv:1610.01644*, 2016.
- [7] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.